

Success-Based System Development Business Plan

Moving from Man-Month Billing and Labor-Intensive Work to Knowledge-Intensive Work

Transforming the Organization to Create Innovation

March 14, 2018

Hirano System Design Office

Profile

Joined CSK Corporation (now SCSK) and worked on process control at a steelmaking plant of a major steel manufacturer. After leaving CSK, worked mainly as an independent professional in financial services (non-bank finance, credit, securities, and IC cards for railway companies and group companies), covering system planning, management, standardization, and design. Also worked on outsourcing business planning and management, comprehensive system planning and management for a major retailer, system integration associated with municipal mergers, internal-control development, and quality improvement in system-development projects. Served as a part-time instructor at Kyushu Electric Technical College until 2002. Began research on system-development technology and IT evaluation systems in 1992 and has continued the research to the present.

1. Overview of the Success-Based System Development Business

1.1 Business concept

Instead of conventional man-month billing, customer profit improvement resulting from the effects of system investment is used as the basis for billing.

1.2 Background

Japan's international competitiveness had declined relatively, while profitability in the IT industry was lower than in Europe, the United States, and India. Maintaining large-scale systems could become difficult. Rapid transformation toward a digital society would increase the need to consider the balance of IT utilization. Continued contract-based development, multilayer subcontracting, and man-month billing could intensify the effects of demographic aging. Subcontracted engineers could face worsening working environments, employment conditions, and compensation while supporting Japan's IT infrastructure.

1.3 Business need

Perspective	Objective
Customer	Determine the balance of IT investment and increase organizational added value.
Development side	Move from labor-intensive to knowledge-intensive work, establish mechanisms for organizational and individual learning, and transform the organization so that it can generate revenue and innovation.
Industry as a whole	Contribute to the development of industries in Japan and overseas.

2. Objective of the Revenue Structure

Under man-month billing, revenue is proportional to the number of employees; when orders decline, a cash-flow shortage can occur. Under a success-based model, customer profit is related to revenue, so the number of employees and revenue do not have to be proportional.

Revenue structure comparison

The original material presents a conceptual comparison between man-month billing (labor-intensive) and success-based billing (knowledge-intensive), including changes in sales and the relationship between revenue and staffing.

Original Japanese Reference

[Click here to open the original Japanese PDF](#)

The detailed original chart is intentionally retained in Japanese so that its axes, arrows, labels, and relationships are not altered by translation.

3. Overview of the Success-Based System Development Business Process

The original process diagram compares conventional contract development with the success-based approach. The success-based approach adds planning and verification of customer profit contribution, investment recovery, gain sharing, and risk management before and after development.

Original Japanese Reference

[Click here to open the original Japanese PDF and view the original process diagram](#)

4. Contract Concept and Verification Policy Applying Game Theory and Contract Theory

4.1 Contract concept

To share the maximized gains, prior discussions and a binding agreement are required. A fair settlement point in an alliance involving a disagreement point is described using four conditions: efficiency, symmetry, invariance, and independence of irrelevant alternatives.

Gain-sharing calculation: Development/alliance-side gain = disagreement gain (reference point / investment amount) + α (negotiation conditions / contribution) < total gain.

4.2 Effect-verification policy

- Clarify what had been a black box within the company, particularly the roles and target figures of IT and non-IT activities.
- Eliminate information asymmetry.
- Clarify the boundaries of the enterprise.

Original Japanese Reference

[Click here to open the original Japanese PDF for the detailed contract and game-theory figures](#)

5. Overview of Evaluation for Success-Based System Development

Applying performance management and knowledge-creation management; constructing an “Innovation Accounting” framework.

The evaluation framework analyzes the relationship between changes in the general ledger and activities of IT and non-IT operations, including SCM, CRM, and HRM, and changes in profit.

The original figure contains detailed arrows and relationships among strategy, activities, enterprise ontology, accounting, performance-management methods, inference/expert systems, data science, and other elements. To preserve the original meaning exactly, the detailed figure is not redrawn in English.

Original Japanese Reference

[Click here to open the original Japanese PDF and view the complete evaluation figure](#)

The English text provides the surrounding explanation, while the original Japanese figure is used as the authoritative visual reference.

Integrated methodologies include SCM, CRM, ABM, HRM, BSC, RCA, ABC, and ABRP, together with enterprise ontology, inference/expert systems, and data science. The purpose is to amplify organizational capability while keeping people at the center.

6. Essential Objectives and Means for Success-Based System Development

Objectives: High added value · High quality · Short development time · Low cost · Safety

Means	Need	Content
People	Software development depends on trained and educated people appropriate to the purpose; people generate results.	Project Management Office; business planners; software engineers.
Methodologies	Development depends on practical processes based on suitable organizational structures.	Management engineering; contract theory; software engineering; ontology, quality and safety engineering.
Management	Improve how results can be achieved through people.	Evaluation and measurement methods.
Tools	Strengthen human-resource quality, manage work, and systematize work.	Business processes; development; configuration management; testing; deployment; project and operations management; groupware.
Reuse	Achieve low cost, short development time, high productivity, and high quality.	Componentization by process/function; enterprise ontology; design; development; testing.
Re-engineering	Improve understanding of existing systems and promote maintenance automation.	Analysis; restructuring; reverse engineering; migration.

Original Japanese Reference

[Click here to open the original Japanese PDF for the detailed original figures](#)

7. Departmental Planning Overview

An organization in which people grow, have autonomy and purpose, engage in skilled dialogue and empathy, and are supported by finance to create innovation.

Plan	Basic policy
Organizational plan	Form teams by professional role and operate through task-force-style teams. Delegate authority as far as possible to improve efficiency, autonomy, capability, and focus on outcomes.
Education and training plan	Train people to focus on outcomes, support self-development plans, encourage feedback, provide training expenses and time, and learn from failure.
Human-resources system plan	Separate personnel evaluation from compensation evaluation. Personnel evaluation should primarily support individual self-development.
Talent acquisition plan	Recruit people motivated to acquire needed skills quickly, with their own views but flexibility when project completion requires delegation.
Financial plan	Move from an annual-budget cycle to financial forecasting, planning, and reporting. Link funding to business activities and repeat short planning/forecasting/monitoring cycles.
Risk-management plan	Minimize hazards and obstacles, reduce uncertainty while complying with rules, and view risk as a potential business opportunity.

8. People, Organization, and Innovation

The material emphasizes a transition from labor-intensive work to knowledge-intensive work. The organization is intended to amplify human capability through appropriate methods, tools, learning mechanisms, autonomy, dialogue, and outcome-based management.

Human element	Organizational element	Business element
Skills, motivation, learning, dialogue, empathy, leadership	Task-force teams, complementary roles, delegated authority, feedback, culture	Outcome measurement, investment recovery, gain sharing, risk management, innovation

Core direction

People → learning and autonomy → knowledge and innovation → customer outcomes → sustainable revenue structure

9. Closing Perspective

The proposal seeks to change the basis of system development from labor input and man-month calculation toward knowledge, contribution, and measurable customer outcomes. Contract, evaluation, organizational, and technical mechanisms are intended to work together rather than as isolated practices.

The 2018 material therefore presents success-based system development not only as a billing method, but as a broader organizational and management model involving performance evaluation, knowledge creation, human-centered work, and innovation.

Thank you for your attention.

Hirano System Design Office (sole proprietorship)
Naohiko Hirano
Fukuoka, Japan

10. Reference to the Original Japanese Material

For figures whose detailed arrows, relationships, labels, or layout are important to the original meaning, this English edition intentionally refers readers to the original Japanese PDF rather than redrawing the figure in a simplified form.

[Original Japanese Reference — Click to open the Japanese PDF](#)

This approach preserves the original visual information while allowing the explanatory text to be read in English.

Original date: March 14, 2018

English edition prepared for research archive review